

Designing a sustainable-resilient-responsive supply chain network considering uncertainty in the COVID-19 era

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ABSTRACT

Effective supply chain management is crucial for economic growth, and sustainability is becoming a key consideration for large companies. COVID-19 has presented significant challenges to supply chains, making PCR testing a vital product during the pandemic. It detects the presence of the virus if you are infected at the time and detects fragments of the virus even after you are no longer infected. This paper proposes a multi-objective mathematical linear model to optimize a sustainable, resilient, and responsive supply chain for PCR diagnostic tests. The model aims to minimize costs, negative societal impact caused by shortages, and environmental impact, using a scenario-based approach with stochastic programming. The model is validated by investigating a real-life case study in one of Iran's high-risk supply chain areas. The proposed model is solved using the revised multi-choice goal programming method. Lastly, sensitivity analyses based on effective parameters are conducted to analyze the behavior of the developed Mixed-Integer Linear Programming. According to the results, not only is the model capable of balancing three objective functions, but it is also capable of providing resilient and responsive networks. To enhance the design of the supply chain network, this paper has considered various COVID-19 variants and their infectious rates, in contrast to prior studies that did not consider the variations in demand and societal impact exhibited by different virus variants.

1. Introduction

The intense competition in the global market, customer expectations, and technological advancements have changed the infrastructure of supply chains, making them more complex than ever (Alabi & Ngwenyama, 2023). It is evident that strategic decisions in supply chain management incur significant costs and can be challenging to modify. In general, supply chain networks are designed to remain stable over extended periods, despite facing pressing challenges related to cost reduction, flexibility improvement, and sustainability enhancement (Tavana et al., 2022). The design of an efficient supply chain network has the potential to significantly reduce its operational costs, but it may also make it more susceptible to unforeseen events. Natural calamities and human-made disasters, such as floods, earthquakes, pandemics, and so on, can lead to economic, productivity, and operational disruptions

(Cardoso et al., 2023). Therefore, supply chain networks must remain resilient to a variety of disruptions.

Coronavirus (COVID-19) has posed a threat to global health and hampered the global economy, as it has stagnated (Mosallanezhad et al., 2023). The COVID-19 outbreak is not only a global health emergency, it is also a labor market and economic challenge, which is increasing the number of infections (Gamal et al., 2022).

Based on the daily new confirmed cases, Fig. 1 provides a general indication of the COVID-19 spread rate. It is obvious that the number of reported cases is lower than true infections because of the restrictions in testing (Johns Hopkins Coronavirus Resource Center, 2022).

During the pandemic, in addition to the priority of preparing medical equipment, a flexible supply chain network is a critical component. Due to the lockdowns announced by governments and severe restrictions on travel, supply chain disruptions are increasing at an alarming rate

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(Bassiouni et al., 2023). Due to the COVID-19 impact, SCM (See Table 1 for the abbreviations used in this work) of various products has faced new challenges (Montoya-Torres et al., 2021). As pandemics are unpredictable, they are classified as special cases of risk for SCM (Bender et al., 2022). It is very difficult for professionals to predict the effects of the COVID-19 pandemic on supply chains because they can cause long-term problems (Nayeri et al., 2023). In the event of an outbreak, logistics capacity must be identified. Furthermore, to ensure that SCN distribution systems are functional and efficient across global disasters, facilities estimation and optimization, as well as flexible modes of transportation are crucial (Foroozesh et al., 2022).

Another challenge to have a proper supply chain during the COVID-19 pandemic is to tackle such a huge volume through the complicated network of companies, logistics providers, and medical departments located in various countries (Tirkolaee et al., 2022). As an example, vaccine supply chain researchers examine the best strategies to meet the needs of the global population (around 7 billion people). Even if vaccination rates reach 75%, over 12–15 billion doses of vaccination will be required to combat the current epidemic (Rele, 2021). Thus, the COVID-19 is distinct from similar viruses such as the SARS pandemic in 2003, the MERS in 2015, Ebola in 2014 and 2018, and Zika in 2016. The ability of the supply chain to respond to fluctuations in demand, both in terms of volume and product type is highlighted by this distinction (Holweg, 2005). Moreover, the incorporation of dependent demand factors to manage uncertainty risks during COVID-19 presents a challenge for supply chain partners to achieve an operable profit (Alkahtani et al., 2021). As part of the effort to make supply chains more responsive, researchers have called for an improved supply chain network, capable of dealing with the intense disruptions caused by the pandemic in order to face all the challenges during and after the pandemic (Jacobsen, 2020). Due to this, the concept of resiliency for SCN has drawn considerable attention from scholars in the COVID-19 era (Katafuchi et al., 2021) (Ivanov & Dolgui, 2020).

It is essential to detect severe acute respiratory syndromes, such as COVID-19, as early as possible in order to control the current epidemic (Rahman et al., 2021). An acute supply shortage of diagnostic kits was caused by the SARS-CoV-2 pandemic (Jeong et al., 2021). In recent years, the healthcare industry has prioritized pandemic preparedness to mitigate the impact of disease outbreaks and their long-term effects. A sustainable, resilient, and responsive supply chain is critical for ensuring an adequate supply of diagnostic kits. Any weakness in the supply chain

Table 1

List of Abbreviations.

Definition	Abbreviation
Polymerase Chain Reaction	PCR
Supply Chain	SC
Supply Chain Network	SCN
Supply Chain Management	SCM
Linear Programming	LP
Mixed-Integer Programming	MIP
Mixed-Integer Linear Programming	MILP
Sustainable-Resilient-Responsive Supply Chain Network Design	SRRSCN
Goal Programming	GP
Multi-Choice Goal Programming	MCGP
Revised Multi-Choice Goal Programming	RMCGP

network could result in a significant shortage of diagnostic kits, underscoring the importance of a robust and adaptable supply chain.

During a pandemic, a sustainable supply chain can address supply chain disruptions in critical areas (Gupta & Soni, 2021). The supply chain should integrate three aspects, including social, environmental, and economic objectives (Bassiouni et al., 2023). The COVID-19 pandemic has disrupted the sustainable supply chain, resulting in significant negative effects on sales rate, company revenue, customer health, logistics service, and overall performance (Valizadeh et al., 2023).

This paper focuses on developing a sustainable, resilient, and responsive supply chain network (SCN) for polymerase chain reaction (PCR) diagnostic test kits, which are critical during the COVID-19 pandemic. To achieve this, a Linear Programming (LP) model is developed with three main objectives: total cost minimization, reducing the social negative impact of shortages, and minimizing the total carbon footprint. The flow of products between different actors of the SCN has been established to ensure sustainable performance in economic, social, and environmental aspects even in the event of disruptions. The proposed model also emphasizes the importance of resilience in the PCR supply chain network by recommending a contract with backup suppliers, an increase in manufacturing capacity, and the use of various sources.

To ensure responsiveness and customer satisfaction, a minimum demand satisfaction rate is defined as a constraint in all scenarios to fulfill demand over potential demand. The model also considers equity in resource allocation between all demand nodes, as demonstrated in the

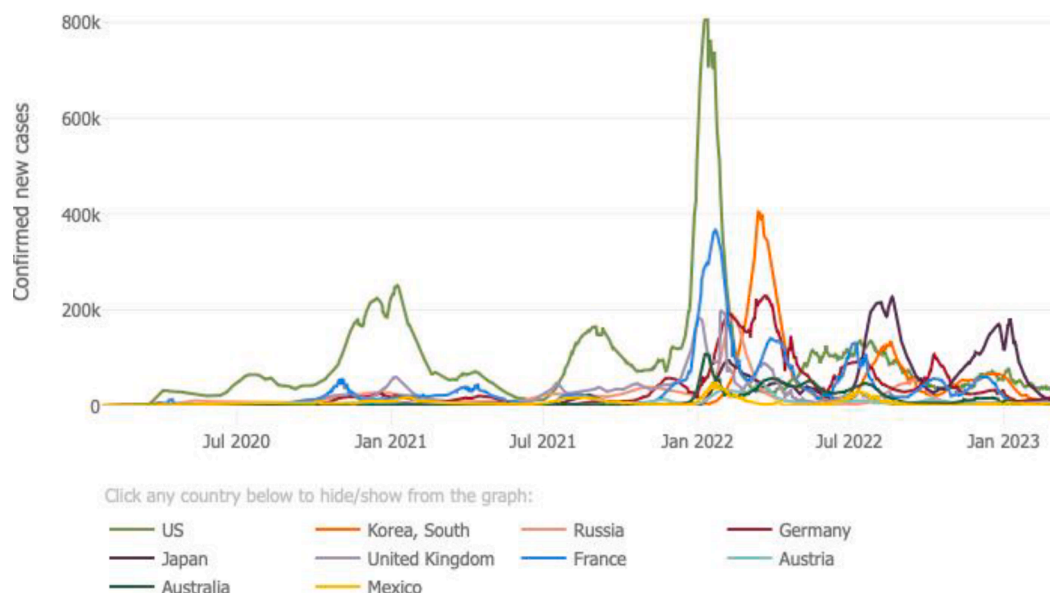


Fig. 1. COVID-19 new confirmed cases per million people per day (Johns Hopkins Coronavirus Resource Center, 2022).

second objective function. To account for uncertain demand conditions, stochastic programming is considered a method for obtaining more realistic results. The model also takes into consideration transportation limitations during the pandemic. The proposed model is highly practical and can be implemented not only for PCR diagnostic test kits but also for any other critical pharmaceutical product with a similar supply network.

This article is organized as follows; Section 2 provides a comprehensive survey related to the research. Section 3 explains and clarifies the methodology used, including the mathematical model, stochastic approach, and linearization. Section 4 discussed the solution method used for the model. Section 5 provides the case description and computational results for the case study. Section 6 analyzes the sensitivity of effective parameters. Section 7 concluded with takeaways, research findings, and further research.

2. Survey on related research

It is undeniable that the pandemic will have catastrophic consequences for the supply chain of various sectors, including agriculture, logistics facilities, transportation, and the healthcare system (Gholian-Jouybari et al., 2023). Public healthcare is characterized by supply chain disruptions that impose a powerful supply structure on health centers. As a result of extraordinary disruptions and increased awareness, people are becoming increasingly concerned about supply chain safety and are demanding sustainable compliance (Fathollahi-Fard et al., 2020). In addition to finding a cure for COVID-19, anticipating trends and developing an effective approach to controlling the disease during the course of a pandemic is also crucial (Tirkolaee et al., 2021). In this section, a precise survey of the literature is provided to spotlight studies relevant to using operations research techniques for SCM of necessary products during the COVID-19 and to identify the research gaps with respect to the current logistics and supply chain issues in healthcare operations.

There have been many studies conducted on different aspects of combating pandemics, especially in the area of logistics and supply chain management. (Büyüktahatkin et al., 2018) suggested a Mixed-integer programming (MIP) model for the logistics operations to control the Ebola epidemic in which they determined the ideal distribution, time, and location of resources. To reduce the overall number of illnesses and fatalities within a constrained budget, a multi-period planning horizon was investigated. Paul & Venkateswaran (2020) developed a generic integrated supply chain epidemic based on robust policies under deep uncertainties to minimize the impact of a pandemic. To find the ensembles of all plausible behaviors of an epidemic, exploratory modeling and analysis methodology were considered. The results highlighted the role of the drug shortage in addition to the shortage duration in controlling the dynamics of an epidemic. Ivanov (2020a) presented a simulation-based analysis approach to monitor and predict the possible effects on supply chain risk during the pandemic. Based on the outcomes from the simulation experiments, the scheduling for facilities' opening/closing can be considered the primary element that influences how the pandemic outbreak affects SC performance. In addition, lead time, the rate of epidemic spread, and the duration of the SC's upstream and downstream disruption are further important variables.

Supply chains that are directly related to the COVID-19 can have very complex networks, and therefore researchers have recently focused on supply chains of vaccines, medical equipment, and face masks. Mosallanezhad et al. (2021) proposed a multi-objective, multi-product, and multi-period model to satisfy the personal protection equipment demands in which the objectives are optimizing total cost and shortage, simultaneously. Alizadeh et al. (2021) suggested a multi-level supply chain for the flu vaccine during COVID-19. Their objectives were minimizing cost, maximization of demand allocation based on lost demand, and customer prioritization.

A multi-objective, multi-level, multi-product, and multi-period

problem was considered by (Goodarzi et al., 2021) to design a sustainable medical supply chain network during COVID-19. Alam et al. (2021) analyzed 15 challenges for the COVID-19 vaccine supply chain and identified the most important challenges. Practical guidelines are suggested for government policymakers to improve the vaccine supply chain for COVID-19. De Sousa Jabbour et al. (2020) organized concepts and trends on resilient and sustainable supply chains in the occurrence of the COVID-19. Guidelines on modifying a more resilient supply chain are presented which highlights an increase in a sustainable consumption perspective. El Korchi (2022) considered a new framework called "Human Needs Supply Chains" which indicates the interdependencies between supply chains, society, and the environment. This study highlighted the links between survivability, resilience, and sustainability in the supply chain. They emphasize that classical resilience is not appropriate to deal with world's challenging and long-term disruptions including COVID-19 and suggest a new supply chain resilience concept to integrate a survivability capacity. Additional research in supply chain network design has been conducted by (Abedsoltan et al., 2022; Bajgani et al., 2022).

The COVID-19 caused supply chain shocks and transformations which indicates the necessity of preserving flexibility and resilience in the supply chain during the current pandemic (Golan et al., 2020). Tirkolaee et al., (2022) proposed a multi-objective MILP model for a multi-period multi-echelon multi-product supply chain. They developed a sustainable study to minimize the total cost, total pollution, and total human risk simultaneously. A stochastic optimization model to design a resilient supply chain operating under random disruptions is recommended by (Sabouhi et al., 2020) and solved by a multi-cut L-shaped solution approach. They determined sourcing and network design decisions to minimize the total cost and ensure that the minimum customer service level is achieved. Ivanov (2020b) analyzed the interconnected supply chain, taking into account survivability and resilience in the context of COVID-19. Results indicated that to guarantee long-term survivability effects, particularly in unusual situations, networks that are highly interconnected and resilient are essential.

During the pandemic, to improve the responsiveness of the supply chain, some research improved logistics by redesigning production facilities and diversifying the location of emergency items, especially personal and protective equipment. A multi-echelon, multi-period responsive SCN developed by (Rabbani et al., 2018) allowed lateral transshipments among retailers that concept of responsiveness was considered based on the delivery times. Singh et al. (2021) considered a simulation model of the public distribution system to explain disruptions during the outbreak. They developed a resilient and responsive food supply chain model to balance the different demands and assist in re-routing vehicles' decision-making. Azaron et al. (2021) proposed a multi-objective two-stage model for a responsive supply chain network under uncertainty to reduce overall trip times while increasing overall profit. They determined the measurement of responsiveness based on total traveling time. Nayeri et al. (2022) designed a supply chain network related to the blood bank refrigerator by considering sustainability, resiliency, as well as responsiveness in a globalized condition. A multi-objective novel model was proposed to minimize the environmental impacts, total costs associated with the network, and to maximize considered social impacts. A modified fuzzy robust stochastic method is considered to deal with the uncertainty. Vali-Siar & Roghanian (2022) developed a multi-objective MILP model for a sustainable-resilient-responsive mixed open and closed-loop SCN. Considering uncertainty, a hybrid robust-stochastic optimization approach is proposed. In addition, the Lagrangian relaxation approach and a constructive heuristic algorithm are considered to overcome the problem convolution and to solve large-scale instances. More studies regarding recent finding on COVID-19 can be found in (Rai et al., 2022; Solayman et al., 2023). Table 2 illustrates the distinctions in various features considered for supply chain networks among the papers analyzed. As depicted, this paper's contributions stand out in comparison to previous works.

Table 2
Summary of papers related to the research.

Research	SC characteristics			Sustainability functions			Scenario-based	Covid-19 characteristics assumptions	Disruption	Equity	Uncertainty approach	Solution method	Case study
	S	R	RP	Econ	SI	Env							
(Sabouhi et al., 2021)	✓	✓		✓	✓	✓	✓				SP, R	Benders decomposition	Downstream petrochemical industries Tire
(Mehrerjerdi & Shafiee, 2021)	✓	✓		✓	✓	✓					SP	Exact	
(Mosallanezhad et al., 2021)				✓			✓	✓	✓	✓	–	MH	PPE
(Hasani et al., 2021)	✓	✓		✓		✓	✓				SP, R	MH	Medical device
(Goodarzian et al., 2021)	✓			✓	✓	✓	✓	✓		✓	Simulation approach	MH	Medicine
(Sazvar et al., 2021)	✓	✓		✓	✓	✓	✓	✓	✓		F	Exact	Vaccine
(Gholami-Zanjani et al., 2021)	✓	✓		✓		✓	✓		✓		SP	Benders decomposition	Food
(Foroozesh et al., 2022)	✓	✓		✓		✓			✓		F, R	Epsilon-constraint	Food
(Vali-Siar & Roghanian, 2022)	✓	✓	✓	✓	✓	✓	✓		✓		SP, R	Lagrangian relaxation, H	Tire
(Nayeri et al., 2022)	✓	✓	✓	✓	✓	✓	✓		✓		F, R	ALWTC	Medical device
(Tirkolaee et al., 2022)	✓			✓	✓	✓	✓	✓			–	NSGA-II	Mask
Current paper	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	SP	RMCGP	PCR test

S: Sustainability, R: Resiliency, RP: Responsiveness, Econ: Economic, SI: Social Impact, Env: Environmental effect, SP: Stochastic Programming, R: Robust, F: Fuzzy, H: Heuristic, MH: Meta-heuristic, ALWTC: Augmented Lexicographic Weighted Tchebycheff, NSGA-II: Non-Dominated Sorting Genetic Algorithm II.

2.1. Research gap

In light of the potential for undeniable disruption in supply chains, the literature review illustrates the importance of paying attention to assorted aspects of SCM problems during a pandemic. Based on reviewed articles, there have been several research works on the SCN problems concerning sustainability, resilience, and responsiveness from various perspectives with a focus on pandemic issues. However, to date, there has not been a comprehensive study that specifically focuses on the development of a model capable of examining the problem of sustainable, resilient, and responsive supply chain network design (SRRSCN) for critical pharmaceutical products, such as diagnostic kits for COVID-19.

As a result of the literature review, it can be concluded that facing an outbreak involves a variety of dimensions, including economic, environmental, and social issues. As a result, neglecting the evaluation of any of these aspects can result in a reduction in the effectiveness of the proposed approach. Researchers have concentrated on multi-objective programming due to the significance of various points of SCM problems during pandemics.

Flexible SCM plays an important role in high-risk pandemics and societies with high costs. Consequently, supply network problems are associated with both private and governmental organizations. The price of diagnostic kits is high, and the lack of a well-designed supply chain can result in additional costs for both suppliers and customers. The authors paid particular attention to various types of costs when designing the supply chain network since cost is always one of the most challenging factors to consider. Moreover, greenhouse emissions are investigated as an important environmental index. There are not many papers with a focus on the environmental problems caused by CO₂ emissions. However, the role of transport-related environmental pollution in a supply chain is irrefutable because the distances intended for transport are usually very long. Especially, in this case, where the transport is done internationally the amount of pollution produced in transportation procedure is significant. The third objective function of the model is to reduce environmental pollution caused by CO₂ gas to

reduce destructive gases.

This paper makes a significant contribution by recognizing the importance of social impact during a pandemic. This includes addressing ethical challenges, preserving trust, ensuring privacy, fairly allocating resources among all demands, demonstrating flexibility in policies to benefit service users, and maintaining emotional control to make rational decisions. The absence of PCR kits, in addition to their crucial role in controlling the pandemic, has undeniable social consequences. It can lead to an increase in infection and death rates, which affects the necessary steps toward achieving recovery (Osofsky et al., 2020). In addition, this problem has long-term aftermath which can cause rising the unemployment rate and depression.

This research was motivated by the need to develop a sustainable resilient-responsive model for a diagnostic kit supply chain that is most effective for sourcing and network design decisions. We propose a model that minimizes the total cost of the PCR test kits shortage as well as the overall negative impact on the environment associated with it.

3. Methodology

This section comprehensively illustrates the suggested methodology to establish a competent sustainable resilient-responsive supply chain for test kits during the COVID-19 pandemic. Fig. 2 depicts a graphical illustration of the supply chain network of PCR test kits. It indicates that four types of suppliers namely, primary suppliers, backup suppliers, international manufacturers, and humanitarian world aids from other countries are included to provide diagnostic kits. In addition, distributors, and clinical centers (e.g., laboratories and hospitals) are considered.

Based on Fig. 2, in this problem quantities of kit supplies are shipped from various suppliers to distribution centers with respect to demands. These facilities are responsible for distributing products among demand nodes. Then, clinical centers receive their supply from distribution centers. Considering time, carbon emission, and cost of transportation, the chosen type of vehicle for shipment might be different. It is important to recognize that the availability of humanitarian aid varies across

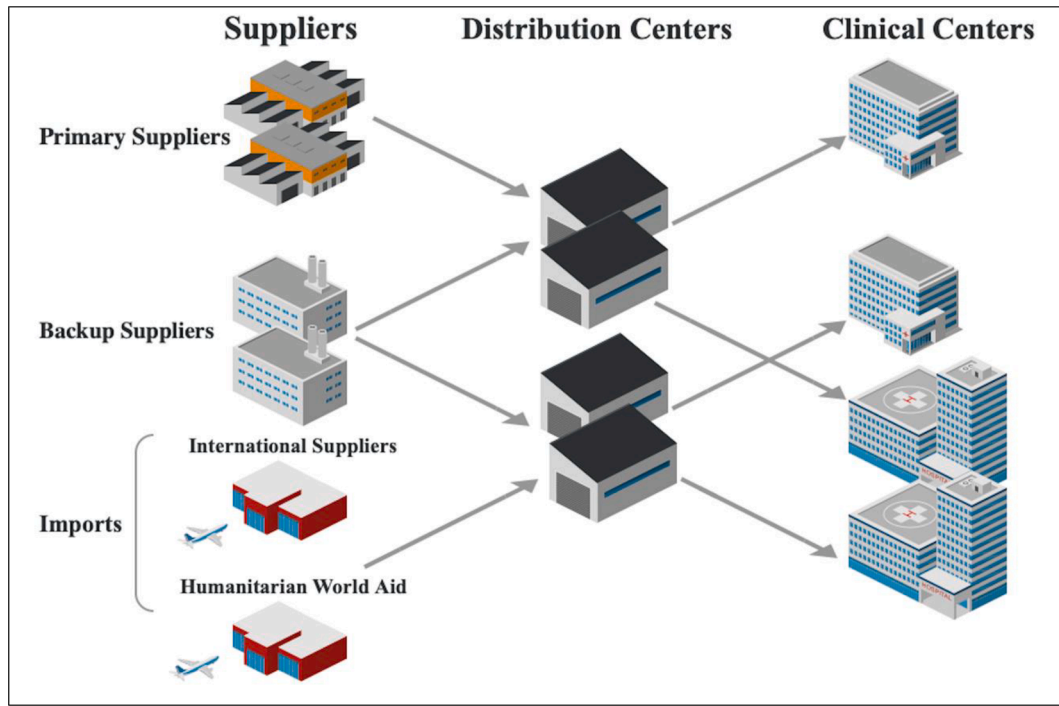


Fig. 2. The supply chain network of PCR test kits.

different scenarios due to the potential challenges that countries may face in supplying aid for themselves. Furthermore, suppliers, distribution centers, and clinical centers exhibit heterogeneity in their respective capacities for supply and demand. This heterogeneity also applies to backup suppliers and their supply capacities. To optimize the SCM and design an SRRSCN, several assumptions are considered which are explained in the following section and Fig. 3 illustrates the methodology of this research.

3.1. Problem characterization and mathematical model

The main problem's objectives and assumptions are described as follows.

- A comprehensive SRRSCN problem is studied considering all three important features of sustainability, resiliency, and responsiveness, simultaneously.
- Multiple national and international sources are considered as suppliers.
- Distribution centers are capacitated.
- Multiple modes of transportation are considered.
- Maximum distance coverage for each vehicle is assumed.
- For each vehicle, different costs, specific capacity, and carbon emissions are considered.
- The possibility of capacity extension for suppliers and distribution centers is assumed.
- Each clinical center is associated with a specific demand.
- Different types of clinical centers are considered, based on their demand.
- The social impact rate for the COVID-19 is considered.

The suggested model's mathematical notations, which include sets and indices, parameters, and decision variables, are listed below.

Sets and indices

S	Set of disruption scenarios, $s \in S$
P	Set of primary suppliers
B	Set of backup suppliers
G	Set of international suppliers
H	Set of humanitarian world aid suppliers
$N = F \cup D \cup K$	Set of all nodes indexed by $i, j \in N$
$F = P \cup B \cup G \cup H$	Set of all suppliers
D	Set of distribution centers
C	Set of hospitals as clinical center nodes
L	Set of laboratories as clinical center nodes
$K = C \cup L$	Set of all clinical centers
A	Set of airplane vehicles as transportation type I
T	Set of truck vehicles as transportation type II
$V = A \cup T$	Set of all transportation modes $v \in V$

Parameters

FC_i	Fixed cost of installing backup supplier $i \in B$
PCE_i	Production capacity extension cost for supplier $i \in P$
UC_i	Cost per unit for each kit produced by supplier $i \in F$ including primary, backup, and international supplier
TC_v	Transportation cost per kilometer for shipping by vehicle v
FTC_v	Fixed transportation cost for operating vehicle v
SC_i	Fixed cost of import test kits from international supplier $i \in G \cup H$
CE_v	Amount of Carbon emission for operating the vehicle v
CP	The maximum amount of carbon that can be produced during supply operation
Dis_{ij}	The distance between node i and node j
DE_{is}	Demand for kits from clinical center $i \in K$ in scenario $s \in S$
$CAPV_v$	The unit capacity of vehicle v for shipping test kits
$LDIS_v$	The maximum distance coverage for each vehicle $v \in V$
$MinDIS$	The minimum distance allowance to operate transportation type I
$CAPS_i$	The amount of supply capacity for supplier $i \in P \cup B$
$CAPL_{is}$	The amount of supply capacity for international supplier $i \in G \cup H$ in scenario $s \in S$
ES_{is}	The maximum amount of extended capacity of supplier $i \in P$ in scenario $s \in S$

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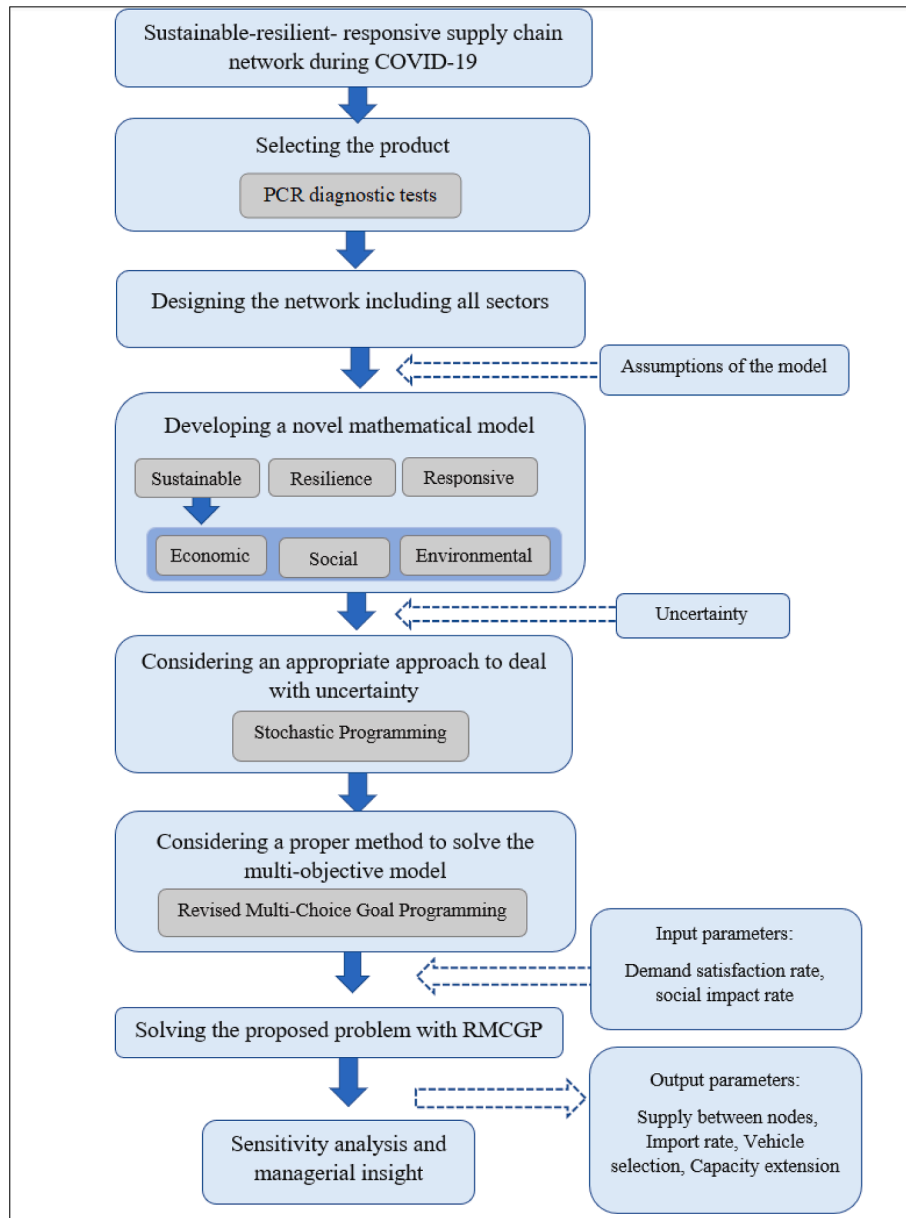


Fig. 3. The methodology of the study.

Parameters (continued)

$CAPD_i$	The capacity amount for the distribution center $i \in D$
$ECAP_{is}$	The maximum amount of extended capacity of distribution center $i \in D$ in scenario $s \in S$
DCE_i	The cost of extension capacity of distribution center $i \in D$
IR_i	Import rate showing the total supply which can be imported from international supplier $i \in G \cup H$ per shipment
SI_s	Social impact rate of the COVID-19 in scenario $s \in S$
AV_{vs}	Availability of vehicle $v \in V$ in scenario $s \in S$
M	A positive large enough number
α_s	The minimum allowed percentage of demand satisfaction in scenario $s \in S$

Decision variables

Φ_{is}	A binary variable equals to 1 if primary Supplier $i \in P$ chooses to extend the production capacity in Scenario $s \in S$; 0, otherwise (o.w).
K_{is}	A binary variable equals to 1 if Distribution center $i \in D$ chooses to extend its capacity in Scenario $s \in S$; 0, o.w.

(continued on next column)

Decision variables (continued)

Π_{is}	Total number of imports from international supplier $i \in G \cup H$ in scenario $s \in S$
Δ_{is}	Quantity of test kits shipped from supplier/distribution center $i \in F \cup D$ to distribution center/clinical center $j \in D \cup K$ by vehicle v in scenario $s \in S$; 0, o.w.
X_{ijvs}	A binary variable is equal to 1 if the distance traveled from node i to j is high enough to operate airplane for shipping; 0, otherwise.
ρ_{ij}	A binary variable equals to 1 if Vehicle v is chosen to ship product from Supplier/Distribution center $i \in F \cup D$ to Distribution center/Clinical center $j \in D \cup K$ in Scenario $s \in S$; 0, o.w.
Λ_{ijvs}	A binary variable which is 1 if primary Supplier $i \in P$ chooses to extend the production capacity in Scenario $s \in S$; 0, o.w.

3.1.1. Objective functions

The following mathematical model best describes the given supply network for PCR test kits represented in the previous section.

$$\begin{aligned} MinZ1_s = & \sum_{i \in B} FC_i * \Phi_{is} + \sum_{i \in P} PCE_i * K_{is} + \sum_{i \in D} DCE_i * \Pi_{is} \\ & + \sum_{v \in V} \sum_{j \in N} \sum_{i \in N} TC_v * Dis_{ij} * X_{ijvs} + \sum_{v \in V} \sum_{j \in N} \sum_{i \in N} FTC_v * \Lambda_{ijvs} \\ & + \sum_{v \in V} \sum_{j \in D} \sum_{i \in F} UC_i * X_{ijvs} + \sum_{i \in G \cup H} SC_i * \Delta_{is} \end{aligned} \quad (1)$$

$$MinZ2_s = \max_i (SI_s * (DE_{is} - \sum_{v \in V} \sum_{j \in D} X_{jivs})) \quad (2)$$

$$MinZ3_s = \sum_{v \in V} \sum_{i \in N} \sum_{j \in N} Dis_{ij} * CE_v * \Lambda_{ijvs} \quad (3)$$

In this study, three objective functions are taken into account. Equation (1) as the first objective function states minimization of the total cost of supply chain operation per scenario including the cost of backup supplier installation, cost of production capacity extension implemented by primary suppliers, capacity extension cost of distribution centers, transportation cost per unit and per vehicle, the total cost of PCR test kits bought for test execution, and import-related costs. Equation (2) as the second objective function indicates the minimization of the social negative impact of shortage for each scenario in which, inequity in supplies' distribution between clinical centers leads to a catastrophe that affects the whole community. Therefore, it is critical to minimize the maximum amount of shortage among all the clinical centers. Consequently, this shortage causes more untested infected people who can transmit the virus at a constant rate based on the variant of COVID-19, and this transmission finally leads to death because of infection. Equation (3) considers the minimization of the total carbon footprint produced in the process of the supply chain specifically during transportation.

3.1.2. Constraints

$$\sum_{j \in D} \sum_{v \in V} X_{ijvs} \leq (CAPS_i + ES_{is} * K_{is}) \quad \forall i \in P, \forall s \in S \quad (4)$$

$$\sum_{j \in D} \sum_{v \in V} X_{ijvs} \leq CAPS_i * \Phi_{is} \quad \forall i \in B, \forall s \in S \quad (5)$$

$$\sum_{i \in F} \sum_{v \in V} X_{ijvs} \geq \sum_{i \in K} \sum_{v \in V} X_{jivs} \quad \forall j \in D, \forall s \in S \quad (6)$$

$$\sum_{i \in F} \sum_{v \in V} X_{ijvs} \leq (CAPD_j + ECAP_{js} * \Pi_{js}) \quad \forall j \in D, \forall s \in S \quad (7)$$

$$\sum_{i \in F} \sum_{j \in D} X_{ijvs} \leq CAPV_v \quad \forall v \in V, \forall s \in S \quad (8)$$

$$\sum_{i \in D} \sum_{j \in K} X_{ijvs} \leq CAPV_v \quad \forall v \in V, \forall s \in S \quad (9)$$

$$X_{ijvs} \leq M * \Lambda_{ijvs} \quad \forall i, j \in N, v \in V, \forall s \in S \quad (10)$$

$$\sum_{i \in F} \sum_{j \in D} \Lambda_{ijvs} \leq 1 \quad \forall v \in V, \forall s \in S \quad (11)$$

$$\sum_{i \in K} \Lambda_{jivs} \leq 1 \quad \forall v \in V, \quad \forall j \in D, \forall s \in S \quad (12)$$

$$\sum_i \sum_j \Lambda_{ijvs} * Dis_{ij} \leq LDIS_v \quad \forall v \in V, \forall s \in S \quad (13)$$

$$\frac{\sum_{j \in D} \sum_{v \in V} X_{jivs}}{DE_{is}} \geq \alpha_s \quad \forall i \in K, \forall s \in S \quad (14)$$

$$\sum_{j \in D} \sum_{v \in V} X_{ijvs} \leq IR_i * \Delta_{is} \quad \forall i \in G, H, \forall s \in S \quad (15)$$

$$IR_i * \Delta_{is} \leq CAPI_{is} \quad \forall i \in G, H, \forall s \in S \quad (16)$$

$$\sum_{j \in N} \sum_{i \in N} \Lambda_{ijvs} \leq M * AV_{vs} \quad \forall v \in V, \forall s \in S \quad (17)$$

$$Dis_{ij} + M * (1 - \rho_{ij}) \geq MinDIS \quad \forall i, j \in N \quad (18)$$

$$Dis_{ij} - M * \rho_{ij} \leq MinDIS \quad \forall i, j \in N \quad (19)$$

$$\Lambda_{ijvs} \leq \rho_{ij} \quad \forall i, j \in N, v \in A, \quad \forall s \in S \quad (20)$$

$$X_{ijvs}, \Delta_{is} \in \mathbb{Z}^+ \quad \forall i, j \in N, v \in V, \quad \forall s \in S \quad (21)$$

$$\Phi_{is}, K_{is}, \Pi_{is}, \Lambda_{ijvs}, \rho_{ij} \in \{0, 1\} \quad \forall i, j \in N, v \in V, \forall s \in S \quad (22)$$

Equations (4) and (5) examine the maximum quantity of PCR test kits that primary and backup suppliers can supply. Equation (6) states the balance between the number of kits received and sent by distribution centers. Equation (7) indicates the maximum quantity of kits that distribution centers can manage in operation based on labor work, warehouse space, and other required facilities. Equations (8) and (9) are the vehicle capacity constraints for carrying kits. Equation (10) notes that the shipment of test kits from one node to another by a vehicle depends on operating that vehicle. Equations (11) and (12) assign each vehicle to each shipment. Equation (13) indicates that the total distance traversed by an individual vehicle should not exceed the specified amount due to the time constraints in healthcare supply chain operations. Equation (14) guarantees a minimum demand satisfaction rate considered for all clinical centers. Equations (15) and (16) emphasize the maximum capacity of kits that can be shipped by import from international suppliers and humanitarian world aid. Equation (17) states the availability of different types of shipment vehicles during disruption operations. Equations (18)-(20) indicate that airplane-type vehicles are used only for specific distances. The implication is that airplanes are not used for supply operations unless the distance between the origin and destination exceeds a certain range. Equations (21) and (22) determine the decision variables' signs and types.

3.2. Stochastic programming

The supply chain network needs to remain flexible in uncertain circumstances in order to remain reliable and responsive. As a general rule, uncertainty refers to the difference between the information that is needed to complete a task and the information that is available (Galbraith, 1974). There have been a number of approaches proposed to facilitate decision-making in uncertain circumstances due to the considerable role uncertainty plays in the operation of a supply chain. The most practical approaches for dealing with uncertain situations are stochastic programming, fuzzy programming, and robust optimization (Goli et al., 2021). Stochastic programming is appropriate when historical information demonstrating parameter behavior is accessible and manifested by probability distribution functions. For parameters with characteristics of fuzzy numbers, fuzzy programming is used (Zhang et al., 2022; Zhang et al., 2020). Robust programming is usually considered when sufficient data is not available to estimate the probability distribution function of the aforementioned uncertain parameters (Ben-Tal et al., 2009).

In this study, to address the existing uncertainty in parameters, the stochastic programming approach is being used. The implemented stochastic programming method is following the one provided by (Aghezzaf et al., 2010). Besides, the min-max relative regret approach proposed by (Inuiguchi & Sakawa, 1995) is also considered. According to (Inuiguchi & Sakawa, 1995), the largest discrepancy between the objective function's best value across all scenarios and each scenario's best value is indicated in mentioned min-max regret method. Notations

of solution used for the model are provided below.

π_s	Probability of occurrence of individual scenario s
Zo_s	Objective function o 's value considering occurrence of scenario s
η_o	The coefficient factor is bounded by the maximum regret cost of objective function o and optimal point of objective function o
Zo_s^*	Objective function o 's optimal value while scenario s happens

The following mathematical model concerning stochastic programming method has been used:

$$\text{Min} \sum_s (\pi_s \times Z1_s) + \eta_1 \times \max_{s \in S} \left(\frac{Z1_s - Z1_s^*}{Z1_s^*} \right) \quad (23)$$

$$\text{Min} \sum_s (\pi_s \times Z2_s) + \eta_2 \times \max_{s \in S} \left(\frac{Z2_s - Z2_s^*}{Z2_s^*} \right) \quad (24)$$

$$\text{Min} \sum_s (\pi_s \times Z3_s) + \eta_3 \times \max_{s \in S} \left(\frac{Z3_s - Z3_s^*}{Z3_s^*} \right) \quad (25)$$

Part one of equation (23) indicates the associated cost of supply chain network under various scenarios, while the second term shows the risk corresponding to diverse results. This equation minimizes the network's related costs as well as minimizing the maximum possible difference that might occur between the suitable best value and the objective function's optimal value considering the occurrence of individual scenarios. Equation (24) represents the network's social impact and its proportionated risk, which minimizes the maximum shortage among all the clinical centers operating and minimizes the second term explained before. Equation (25) quantifies the network's environmental impacts and its corresponding risk, which minimizes the total carbon footprint produced in the process of the supply chain and the maximum difference between the best and optimal objective functions.

3.3. The model linearization

According to the developed model, the second objective function is nonlinear. To simplify the model, this function is linearized, and consequently, equation (26) is considered as the objective function subject to equation (27).

$$\text{Min} Z2_s = TS_s \quad (26)$$

$$SI_s^* (DE_{is} - \sum_{v \in V} \sum_{j \in D} X_{jivs}) \leq TS_s \quad \forall i \in K, \forall s \in S \quad (27)$$

In addition, equations (23)-(25) are nonlinear due to the existence of max function. In the equivalent linear model, these equations are rewritten as follows:

$$\text{Min} Z1 = \sum_s (\pi_s \times Z1_s) + \eta_1 \times REG_1 \quad (28)$$

$$\text{Min} Z2 = \sum_s (\pi_s \times Z2_s) + \eta_2 \times REG_2 \quad (29)$$

$$\text{Min} Z3 = \sum_s (\pi_s \times Z3_s) + \eta_3 \times REG_3 \quad (30)$$

St.

$$\frac{Z1_s - Z1_s^*}{Z1_s^*} \leq REG_1 \quad \forall s \in S \quad (31)$$

$$\frac{Z2_s - Z2_s^*}{Z2_s^*} \leq REG_2 \quad \forall s \in S \quad (32)$$

$$\frac{Z3_s - Z3_s^*}{Z3_s^*} \leq REG_3 \quad \forall s \in S \quad (33)$$

REG_o is the maximum value related to regret of objective function O and defined for linearization of the mentioned equations. Equations (31)-(33) calculate the amount of REG_o while ensuring they are going to be equal to the maximum deviation from the optimal value.

4. Solution method

Different methods are recommended to solve multi-objective problems including the ϵ -constraint method (Fahimnia et al., 2017), LP-metrics (Curtis et al., 2013), multi-choice goal programming (Chang, 2008), and revised multi-choice goal programming (Chang, 2008).

One of the most used decision-making approaches is the Goal Programming (GP) method proposed by (Charnes & Cooper, 1957). GP is an analytical method to address decision-making problems and minimize the unwanted deviation variables of the model's objectives. In other words, the goal of the method is to reduce the deviations between the target values and aspiration levels (Aalaei & Davoudpour, 2016).

In dealing with real-world decision-making problems, GP is extremely useful when considering multiple controversy objectives. Different types of achievement functions lead to various GP variants. The Weighted Goal Programming (WGP), the preemptive function, and the Chebyshev structure, which minimizes the maximum deviation, are three of the most popular types of achievement functions (Romero, 2004). Based on the available data, decision-makers in GP approaches define a conservative baseline aspiration level and resource boundaries. However, their preference may be lower/higher aspiration levels. For solving this problem and accommodating various aspiration levels, (Chang, 2007) introduced a novel modelling approach called the Multi-Choice Goal Programming (MCGP) with multiplicative terms of binary variables. A multi-choice aspiration level under a given constraint is assumed by the MCGP method to avoid underestimating the decision and to achieve the global optimum solution. Because of the complexity of MCGP model, it is difficult to understand by industry parties. Thus, (Chang, 2008) modified the approach and formulated the Revised Multi-Choice Goal Programming (RMCGP) as follows:

Set O	The set of objective functions
$f_o(x)$	The linear function of all values of x for o th goal
$g_{o.min}$, $g_{o.max}$	The minimum and maximum values of aspiration level for objective $o \in O$
y_o	Continuous variable with the minimum value $g_{o.min}$ and the maximum value $g_{o.max}$
w_o	Weight of deviations from the goal for objective $o \in O$
d_o^+ , d_o^-	Positive and negative deviations from the objective's ($o \in O$) aspirational level
e_o^+ , e_o^-	Positive and negative deviation from the maximum or minimum value of aspiration level of objective $o \in O$
α_o	Weight of deviations from the maximum or minimum value of aspiration level for objective $o \in O$

Based on the type of objective functions, the case of RMCGP varies. Case 1 is "the more the better" formulated as Equations (34)-(38):

$$\text{Min} \sum_{o \in O} [w_o^* (d_o^+ + d_o^-) + \alpha_o^* (e_o^+ + e_o^-)] \quad (34)$$

$$f_o(x) - d_o^+ + d_o^- = y_o \quad \forall o \in O \quad (35)$$

$$y_o - e_o^+ + e_o^- = g_{o.max} \quad \forall o \in O \quad (36)$$

$$g_{o.min} \leq y_o \leq g_{o.max} \quad \forall o \in O \quad (37)$$

$$d_o^+, d_o^-, e_o^+, e_o^- \geq 0 \quad \forall o \in O \quad (38)$$

Case 2 is "the less the better" which is presented as Equations (39)-(43):

$$\text{Min} \sum_{o \in O} [w_o^* (d_o^+ + d_o^-) + \alpha_o^* (e_o^+ + e_o^-)] \quad (39)$$

$$f_o(x) - d_o^+ + d_o^- = y_o \quad \forall o \in O \quad (40)$$

$$y_o - e_o^+ + e_o^- = g_{o,min} \quad \forall o \in O \quad (41)$$

$$g_{o,min} \leq y_o \leq g_{o,max} \quad \forall o \in O \quad (42)$$

$$d_o^+, d_o^-, e_o^+, e_o^- \geq 0 \quad \forall o \in O \quad (43)$$

$f_o(x)$ is objective function o , and X indicates the decision vector. In our research, considering minimizations for all three objective functions, case 2 has been chosen for problem solving. The objective of the RMCGP, which is to minimize the weighted sum of deviations from the aspiration level and its minimum value, is represented by Equation (39). Equation (40) represents the positive and negative deviations from the aspiration level. Equation (41) determines the positive and negative deviations from the minimum possible value of aspiration level. It is also guaranteed by Equation (42) that the minimum and maximum value bounds aspiration levels. The type of variables is indicated in Equation (43). To sum up, the model of the RMCGP for our problem is considered below:

$$\begin{aligned} & \text{Min}[w_1^*(d_1^+ + d_1^-) + \alpha_1^*(e_1^+ + e_1^-) + w_2^*(d_2^+ + d_2^-) + \alpha_2^*(e_2^+ + e_2^-) \\ & + w_3^*(d_3^+ + d_3^-) + \alpha_3^*(e_3^+ + e_3^-)] \end{aligned} \quad (44)$$

$$f1 - d_1^+ + d_1^- = y_1 \quad (45)$$

$$y_1 - e_1^+ + e_1^- = g_{1,min} \quad (46)$$

$$g_{1,min} \leq y_1 \leq g_{1,max} \quad (47)$$

$$f2 - d_2^+ + d_2^- = y_2 \quad (48)$$

$$y_2 - e_2^+ + e_2^- = g_{2,min} \quad (49)$$

$$g_{2,min} \leq y_2 \leq g_{2,max} \quad (50)$$

$$f3 - d_3^+ + d_3^- = y_3 \quad (51)$$

$$y_3 - e_3^+ + e_3^- = g_{3,min} \quad (52)$$

$$g_{3,min} \leq y_3 \leq g_{3,max} \quad (53)$$

$$d_1^+, d_1^-, e_1^+, e_1^-, d_2^+, d_2^-, e_2^+, e_2^-, d_3^+, d_3^-, e_3^+, e_3^- \geq 0 \quad (54)$$

The weighted positive and negative deviations from the aspiration levels and their lower bounds are minimized by Equation (44). Equations (45)-(53) indicate the negative and positive deviations from corresponding objectives and the minimum value which also ensures that aspiration levels are bound to their minimum and maximum values. The characteristics of the variables are shown in Equation (54).

5. Case study

The proposed problem involves a supply chain that has three levels, suppliers, distribution centers, and clinical centers. Using the model, the

supply chain is optimized from an economic, social, and environmental perspective. The assumptions regarding the real condition of the COVID-19 pandemic in Iran are selected as part of the evaluation of our mathematical model under real-world conditions.

On February 19, 2020, Qom reported the first death caused by COVID-19, and thereafter, all other provinces in Iran were affected. As of April 21, 2020, reports illustrate that of 330,137 tested patients, 80,868 people have been infected of which almost 3513 people suffered a critical illness and 5031 people have died (Salimi et al., 2020). Fig. 4 reveals the confirmed COVID-19 cases in Iran from the start of the pandemic. In Iran, the considerable number of confirmed COVID-19 cases negatively influenced each of the economic, environmental, and social aspects.

There has been a dramatic increase in the number of suspected cases of COVID-19 in various provinces during the first week, overflowing hospitals with the disease. Consequently, medical personnel face shortages of protective equipment, life-saving drugs, and treatment facilities due to the panic buying of masks, gloves, and sanitizers. Furthermore, public anxiety is increased by fake news and misinformation. Measures like preventing large-scale gatherings, closing educational facilities, national screening programs, closure of religious and sacred places, and social distancing caused economic, social, and environmental problems in Iran (Zand & Heir, 2020). All these challenges emphasize the role of attention to economic, social, and environmental aspects of a supply chain to tackle unpredictable situations, especially during a pandemic.

5.1. Case assumption

The COVID-19 pandemic has resulted in considerable challenges for Mazandaran, a northern province of Iran. Due to some special geographical features of Mazandaran province, the infection and mortality rate of corona in this region increased rapidly (Kordi et al., 2023; Kordi et al., 2022). Therefore, to attend to the problems in this region and forecast future pandemics, the cities covered by Babol University of Medical Sciences in Mazandaran province were considered as a case study. Three large cities are included in these sections, namely Babol, Babolsar, and Fereydown Kenar.

As stated in the research gap, the supply chain of PCR tests plays a significant role in the control of pandemics, and due to the challenges in the case study area, this importance has doubled. First, according to

Table 3
Scenario description.

Variant's name	Start date in Iran	Possible variant evolving date	Pandemic status
COVID-19	22 February 2020	30 April 2020	Start of the pandemic
Delta	11 May 2021	21 July 2021	Start of the vaccination
Omicron	19 December 2021	24 February 2022	Global vaccination

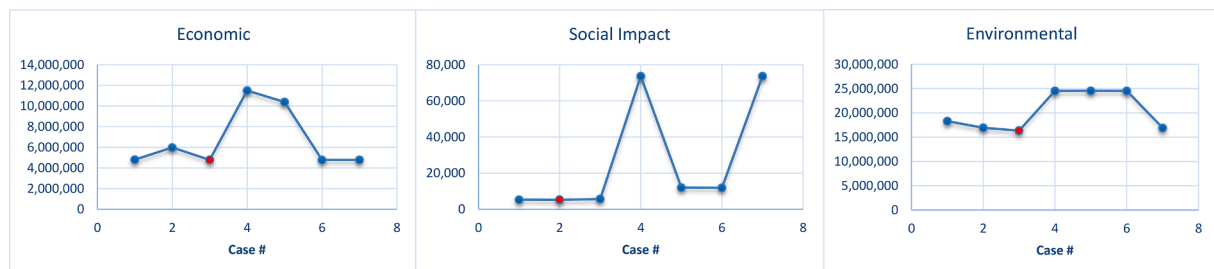


Fig. 4. Daily new confirmed COVID-19 cases in Iran per million people (Johns Hopkins Coronavirus Resource Center, 2022).

different COVID-19 variants and for better analysis, by examining the records of the infection rate considering population, three scenarios have been considered according to the pandemic situation. Table 3 provides information on each scenario.

In general, the developed model can be applied to all products that do not require any special considerations, such as cold chain logistics, that could potentially impact the supply operations. The supply chain of PCR tests in this region includes suppliers, distribution centers, and treatment clinics, the information in each section is described as follows:

In addition to primary suppliers, backup suppliers, international suppliers, and humanitarian suppliers are also intended. From the beginning of the pandemic, three distribution centers in the case study area produced PCR diagnostic tests. Backup suppliers changed with a certain amount of initial cost and establish a PCR test production line. Five backup suppliers meet the needs of the region for PCR testing. During the outbreak, Iran imported PCR tests from different countries to be able to meet the demand. Russia and China are the main suppliers in the case study region. In addition, the World Health Organization, as a humanitarian supplier, sends PCR tests to Iran that part of which send to Mazandaran. “Hejrat” and “Razi” companies are considered the two main distribution centers of PCR testing in Mazandaran, which directly provide the demands of the 12 clinical centers (6 laboratories and 6 hospitals) in the case study area.

The main information of the case study is:

- (1) The case study is constructed as a single-period problem for one day.
- (2) Different types of suppliers are considered, and each type of them has its cost and capacity. The costs and capacities are considered based on the available information of the clinical centers, distribution centers, and suppliers.
- (3) Although various small distribution centers can provide PCR kits in the Mazandaran province, two main centers that provide PCR kits are considered the problem's distribution centers.
- (4) Demand for each clinical center is considered as an estimation of the possible demand proposed by five experts from the waste management department in Babol (Mazandaran University of Medical Sciences, 2021).
- (5) For different scenarios, various allowed minimum percentages of demand satisfaction (α_s) is assumed. The amounts of α_s for COVID-19, Delta, and Omicron are 0.5, 0.55, and 0.2 respectively. The difference between the α_s for COVID-19 and Delta was not considerable. As a result of global vaccination, the COVID death rate decreased and the importance of satisfying all the demands were reduced. Therefore, for the Omicron variant, the risk of disease for the society decreased so α_s is considered lower than other scenarios.
- (6) To give a standard mechanism to monitor the spreading of an infectious disease like COVID-19, the basic reproductive number or R_0 was considered as a metric (Al Zobbi et al., 2020). The average number of susceptible individuals that each diseased person is anticipated to infect is known as the rate of R_0 . Thus, the greater R_0 , the more contagious the disease (Hussein et al., 2021). Less than one value for R_0 indicates that the illness will likely naturally stop spreading and vanish. Generally, the R_0 rate indicates the number of people who are at risk of COVID-19 and these people are vulnerable to COVID-19 from different aspects. For instance, increasing inequality, exclusion, discrimination, and global unemployment in the medium and long term. Therefore, in the case study, the rate of R_0 is considered as the social impact (SI) and shows the number of people who might be endangered socially if a shortage of PCR tests happens.

5.2. Computational results

The real-world case study with the extracted parameters has been

Table 4

Objective functions values to select the upper and lower bound for the aspiration level.

Objective	Z1	Z2	Z3
Z1	2.1×10^6	3.52×10^8	1.8×10^8
Z2	1.12×10^8	1.7×10^3	8.9×10^7
Z3	1.4×10^7	1.08×10^8	7.74×10^6
RMCGP	4.382427×10^7	4.196×10^4	4.876732×10^7

Table 5

Objective function values.

Objective function	Economic	Social Impact	Environmental Effect
Value	4,071.6 (Million Toman)	4732 (person/period)	14.833 (mtCO2)

solved with Intel(R) Core(i7-9700) CPU (3.00 GHz), RAM 16 using the GAMS software to determine the validity of our mathematical model. The solution is obtained by solving each scenario separately to specify the goal, upper and lower limitations of objectives, and the optimal value of each objective in each scenario. Therefore, three individual objective models are used to achieve the best solution structure and assess the model's accuracy. The weights $W_1 = 0.5$, $W_2 = 0.25$, $W_3 = 0.25$, defined for objective functions one to three, respectively. These weights are considered for the RMCGP method based on average values weighted by some experts from the COVID-19 control department in Babol (Mazandaran University of Medical Sciences, 2021). The proposed model is implemented for the case study to attain the optimal solution structure and evaluation of the model's performance. The solving time for the RMCGP method is 42 s, that considered an admissible time for a real-world case during a pandemic. Table 4 shows the value of each objective function and the final value based on the RMCGP method.

Table 5 presents the values of each objective function, including the Economic, Social Impact, and Environmental Effect, along with their respective units.

6. Sensitivity analysis and discussion

Sensitivity analysis is a technique for standardizing the inputs of a statistical model to find critical control points, prioritize gathering more data, and validate the model. It can be a structured method to analyze the impacts of changing critical parameters on the model's output (Karrman & Allaire, 2009). In this section, to evaluate various circumstances and analyze the result of the different situations in the model, the sensitivity of some critical parameters is assessed. Two parameters are considered for analyses and their effects on the results are presented as follows.

6.1. Sensitivity analysis on η_o which is the coefficient factor of the stochastic approach

The use of a stochastic approach involves coefficient factors that are associated with the maximum regret function and deviation illustration. Hence, it is crucial to calibrate these factors accurately to achieve the minimum objective function while considering the deviations of all possible objective values in non-optimal scenarios from their optimal objective values. The selection of the best combination of coefficients that strikes a balance between all objective functions can serve as default coefficients for the given case.

As is shown in Table 6, and Fig. 5, case number 3 shows the best optimal value for coefficient factors in economic and environmental objective functions, and its value in the social impact objective function is close to the optimal value. Therefore, case number 3 has been chosen for all analyses. Results show that the coefficient of zero leads the value

Table 6
Results of sensitivity analysis on the coefficient factors of stochastic approach in various cases.

Case #	η_1	η_2	η_3	Economic objective value				Social impact objective value				Environmental objective value			
				VW*	MRR**	REG ₁	VC***	VW	MRR	REG ₂	VC	VW	MRR	REG ₃	VC
1	0	0	0	4,803,200	0	0.975	4,803,200	5,294	0	1.074	5,294	18,301,300	0	0.742	18,301,300
2	10,000,000	1,000	10,000,000	4,201,300	1,780,000	0.178	5,981,300	4,857	346	0.346	5,203	15,004,940	1,930,000	0.193	16,934,940
3	10,000,000	10,000	10,000,000	4,071,600	710,000	0.071	4,781,600	4,732	940	0.094	5,672	14,833,550	1,510,000	0.151	16,343,550
4	100,000,000	1,000,000	100,000,000	4,003,200	7,500,000	0.075	11,503,200	4,693	69,000	0.069	73,693	14,833,550	9,700,000	0.097	24,533,550
5	100,000,000	100,000	100,000,000	3,990,000	6,400,000	0.064	10,390,000	4,708	7,300	0.073	12,008	14,852,100	9,700,000	0.097	24,552,100
6	10,000,000	100,000	100,000,000	4,071,600	710,000	0.071	4,781,600	4,607	7,300	0.073	11,907	14,833,550	9,700,000	0.097	24,533,550
7	10,000,000	1,000,000	10,000,000	4,092,500	690,000	0.069	4,782,500	4,693	69,000	0.069	73,693	15,268,550	1,640,000	0.164	16,908,550

*Value without considering maximum regret function as the risk associated with deviations of different scenarios, **Maximum regret function, *** Value considering maximum regret function.

of the objective function to fluctuate freely in all three objective functions as is shown in case number 1. On the other hand, considering an abnormal big number for these coefficients as is shown in case number 4 and 5 for all objective functions and case number 7 for the social impact objective, imposes a high objective value since the regret function is fluctuating in an interval of (0.06 to 1) and it's impossible for them to be less than specific amounts.

Furthermore, it is important to point out that the weight of objective functions for economic objective value is two times greater than that of social impact and environmental objective functions. By prioritizing objective functions according to their weights, the model attempts to determine the overall objective value.

6.2. Sensitivity analysis on objective weights of RMCGP

The weights assigned to the objectives in the RMCGP approach are among the most critical parameters that influence the solution. Different weight scenarios have been defined for each economic, social, and environmental aspect to explore how changes in weights impact the value of objectives. To determine the relative importance of each objective, six weight scenarios have been identified based on the expert's opinion. This allows for a comparison of the priority given to each objective under different weight scenarios. Concerning the expert's opinion, the average weights for the RMCGP method in the previous section are considered $W_1 = 0.5$, $W_2 = 0.25$, $W_3 = 0.25$. The maximum weight is assumed for economic aspects which is minimizing the total cost; however, it doesn't imply that this aspect is always more important than other ones. As economic issues can always be influential in real-world problems and consider a key objective, two different scenarios are considered with more emphasis on the cost objective, and the importance of the social objective is assumed half of the economic aspect. In the third scenario, all weights are defined equally in order to analyze a balanced condition. Due to the importance of social challenges and environmental problems during the pandemic, some scenarios are formulated with an emphasis on these factors. Table 7 presents the results. According to the weights associated with each objective, Figs. 6 to 8 illustrate the analysis of each objective.

Fig. 6 demonstrates the effect of W_1 of the RMCGP method on the first objective function which is total cost minimization. Based on Fig. 6, increasing W_1 , the amount of total cost is reduced steadily which indicates the direct effect of W_1 on the first objective. By increasing W_1 , there has been a continuous improvement in cost reduction, so to pay more attention to the economic aspects, the decision-makers can assign more weight to the first objective function. However, there is no significant improvement in costs in the interval of $W_1 = (0.1, 0.4)$, and the graph turns into a discrete line in $W_1 = 0.4$, which shows that there is a correlation between weights. Thus, objective function values do not depend on just their weights. In more detail, in scenario 6, the left $W_1 = 0.4$ in Fig. 6, the environmental aspect is more important, and in scenario 4, the right $W_1 = 0.4$ in Fig. 6, the objective function of social impacts has been given more weight. It can be concluded that social impacts and the total cost are more correlated than environmental effects and total cost. With equal weights for the first objective function, the one with a higher social impact weight leads to greater cost reductions.

Fig. 7 presents an analysis of the social impact objective function based on different scenarios of RMCGP weights. As illustrated in Fig. 7, increasing the weight of W_2 has a minor effect on the social impact objective, and it is almost the same in all different W_2 s less than 0.8. However, based on Table 7 with increasing W_2 the run time is considerably increased. It indicates the sensitivity of social impacts objective function is low. Meanwhile, for the weight of 0.8, the social negative impact reduction is significant, and it demonstrates that the decision-makers can control the improvement of the second objective by increasing the weight of RMCGP.

As shown in Fig. 8, in the first two scenarios the weight is assumed



Fig. 5. Value of each objective function in different case numbers.

Table 7

Scenario description for analyzing RMCGP weights.

Case number	Case	W_1	W_2	W_3	Z_1	Z_2	Z_3	Run time
1	$W_1 > W_2 = W_3$	0.5	0.25	0.25	43,824,270	40,196	48,767,320	41.5
2	$W_1 > W_2 > W_3$	0.6	0.3	0.1	43,824,270	40,196	48,767,320	29
3	$W_1 = W_2 = W_3$	0.33	0.33	0.33	44,213,210	40,196	48,279,520	68
4	$W_1 = W_2 > W_3$	0.4	0.4	0.2	43,824,270	40,196	48,767,320	81
5	$W_2 > W_3 = W_1$	0.1	0.8	0.1	44,255,270	27,693	48,312,420	128
6	$W_1 = W_3 > W_2$	0.4	0.2	0.4	44,213,210	40,196	48,279,520	96

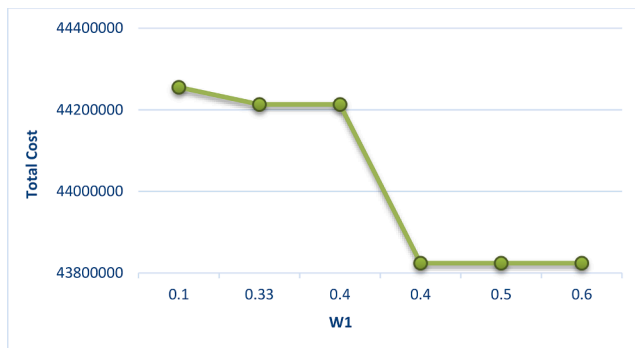


Fig. 6. Analysis of objective function 1 based on W_1 weights of the RMCGP method.

0.1 and the difference is between the weight considered for the first and second objectives. Concerning information on the scenarios in Table 7 and Fig. 8, higher weights for the minimizing social negative impact objective function have some correlations with third objective function optimization. Furthermore, an increase in W_3 by more than 0.25 has a considerable impact on reducing the carbon footprint.

6.3. Sensitivity analysis on demand satisfaction

To evaluate the effectiveness of the proposed model and assess the impact of changes in α , a sensitivity analysis has been conducted on on-demand satisfaction. The satisfaction rates vary depending on the changes in α . In certain scenarios, the demand has responded more than the satisfaction rate based on resource availability, as minimizing negative social impacts takes precedence. Conversely, in other scenarios, the lowest possible satisfaction rate has been considered, based on the priority of the objective functions. These sensitivity analyses allow for a comprehensive evaluation of the model's effectiveness under various conditions. The detail of the scenarios is presented in Table 8 and Fig. 9 indicates the results.

As shown in Table 8 and Fig. 9, in the first scenario, considering various amounts of α the demand satisfaction values remain constant at the maximum value of 1. The supply capacity, extension capacity, and distribution centers allow us to supply more than the demand

satisfaction rate. Therefore, in the first scenario preventing shortage is the highest priority. It means that the objective function of minimization of the social negative impact of shortage has a high priority in the first scenario and demand satisfaction rates are more than α to reduce shortage as possible. This scenario can be very useful in pandemics because shortage costs can cause people's death and accelerate disease spread, so apart from the alpha rate and the satisfaction rate, balancing between the first objective function (economic objective) and the second objective function (social impact) matters.

In both scenarios 2 and 3, the demand satisfaction is in the lowest value because the objective function of total cost minimization and its weight does not allow these values to exceed the minimum α rate, while the difference between these two graphs is the highest amount of supply. The demand satisfaction values are considered the lowest α rate in scenario 2, the highest available supply is 59% for demand satisfaction. However, in the third scenario, the maximum satisfaction rate is 62%. As is evident in Table 8, because of the demand, supply, and distribution center's capacities, it is infeasible to exceed the maximum amount of supply. These scenarios emphasize the impact of objective function weights and demonstrate to give the best results in crises, weighting should be according to priorities. Detailed information on the maximum supply for all the scenarios is specified in Table 9.

Concerning Table 9, in the second scenario, the maximum possible

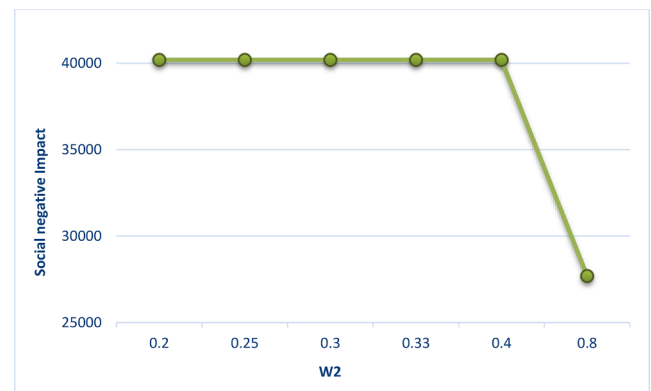


Fig. 7. Analysis of objective function 2 based on W_2 weights of the RMCGP method.

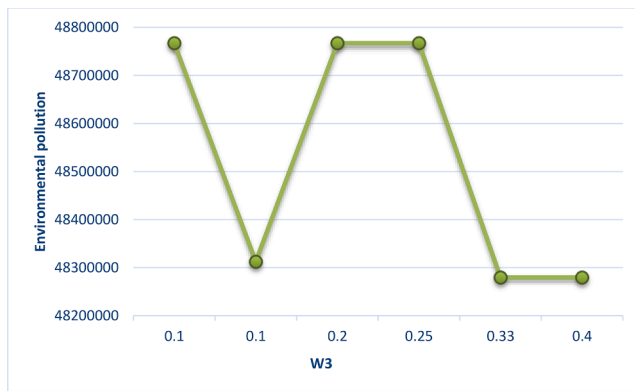


Fig. 8. Analysis of objective function 3 based on W3 weights of the RMCGP method.

Table 8

The detail of each scenario for sensitivity analysis on demand satisfaction.

$\alpha 1$	Satisfaction rate	$\alpha 2$	Satisfaction rate	$\alpha 3$	Satisfaction rate
0.2	1	0.2	0.2	0.2	0.2
0.3	1	0.3	0.3	0.3	0.3
0.4	1	0.4	0.4	0.4	0.4
0.5	1	0.5	0.5	0.6	0.6
0.8	1	0.59	0.59	0.62	0.62
1	1	0.7	Infeasible	0.7	Infeasible

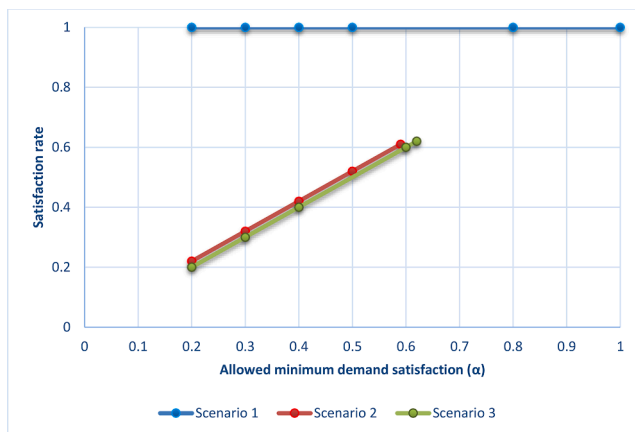


Fig. 9. Sensitivity analysis on demand satisfaction for different scenarios.

supply amount has been done. This model is sufficiently responsive, uses backup suppliers, and satisfies the demands at a meaningful rate, but does not require more distribution centers. For instance, in node 10 distribution centers did not add as it considers unnecessary because the maximum amount of supply has been sent and utilizing more distribution centers is not effective. The third scenario works exactly the opposite of scenario 2, the capacity of distribution centers for nodes 9 and 10 have been extended, however, there isn't enough use of the capacity extension of primary suppliers and utilizing backup suppliers. Only one backup supplier has been used because in this scenario the supply is not a bottleneck. Therefore, the main issue is distribution rather than supply.

6.4. Managerial insights

Considering a case study based on a real-world problem compatible with the characteristic of a world crisis pandemic can provide valuable results in both practitioner and academic areas. Choosing COVID-19 and optimizing a case study based on this pandemic in Iran can benefit

sustainability in all its facets, as managerial parties experienced different challenges in each aspect of economic, social, and environmental. Regarding the research characteristics, it can provide health sectors with management insights pandemic problems in the future. In addition, this research has investigated the PCR supply chain networks problem under uncertainty, therefore, it can give managers a proper perspective on how uncertainty affects the SCN design problem. The results of the proposed model help decision-makers to define an appropriate scenario-based problem based on their preferences that provides the least risk to the economy, society, and environment. Following is a summary of the case study's results and sensitivity analysis' concluded managerial implications based on various perspectives:

- Sustainability:** from an *Economic* aspect, limitation in the budget is always an issue in decision-making. However, in a real-world crisis like COVID-19 which involves a city, a country, and on large scale, the whole world, there are no limitations in supplying critical products like PCR tests. Therefore, though the preference is to spend less on the operation, the responsiveness of the supply network is much more important. Besides, computing the real negative impact on the economy is so sophisticated for countries and probably impossible to do precisely. In the responsiveness issue, there is a question that comes to mind and is "Why don't we use all capacities to reduce the shortage to zero?", in this essence we can't always rely on the ideal assumption, which is the more money we spend, the more capacity of supply will have. Real-world problems like COVID-19, have much more assumptions than just investments. From the *Social impact* aspect, it's been concluded that in scenario-based problems, sometimes dealing with all negative social impacts is not unsolvable and, in some scenarios, capacities are not enough whether in supply, distribution, or other sectors of the network. Besides, computing this negative impact is complicated since, in the context of unmet demand, managers should define a function to relate them. From an *Environmental aspect*, utilizing vehicles with high carbon footprint causes an international crisis of global warming. Though it's unavoidable to use airplanes for transportation of products from long-distance locations, truck delivery and especially for products that are not large in terms of their size can be alternated with a green vehicle like robots and drones (Farajzadeh et al., 2020; Moadab et al., 2022; Song et al., 2022). The environmental cost is not negligible, and the harmful effects of carbon emission of the operation can cause more damage to the economy and involve more people. Therefore, there is a direct relationship between environmental aspects and social impact in the long term.

In conclusion, as long as economic objective functions matter, sustainability also matters since there is a strong connection between the economy, society, and the environment. Since this model was developed to be a linear model, considering non-linear relations for all these three functions could increase the complexity of the model.

- Resiliency:** The importance of resiliency is realized by paying attention to the behavior of the model in various scenarios concerning using backup suppliers and primary supplier capacity extension. We observed that not only extension in capacities matters but also study on the maximum capacity extension matters. In a supply network, targeting bottlenecks is critical and the focus should be on the capacity extension of bottlenecks. Unbalanced capacity between suppliers and distribution centers can cause either unused supply capacity or waste distribution resources.
- Responsiveness:** defining the necessary minimum demand satisfaction rate for a crisis like a pandemic helps to prevent an uncontrollable pandemic. Although results say that it's more probabilistic to see met demand get the value of minimum demand satisfaction rate, however, in some scenarios the supply and distribution capacity

Table 9

Detail of the maximum supply based on the satisfaction rate for each scenario.

α_1	1		α_2	0.59		α_3	0.62	
Satisfaction rate S1	1		Satisfaction rate S2	0.59		Satisfaction rate S3	0.62	
Φ_{i1}	4	1	Φ_{i2}	4	1	Φ_{i3}	4	1
	5	1		5	1		5	–
K_{i1}	1	1	K_{i2}	1	1	K_{i3}	1	–
	2	1		2	1		2	–
	3	–		3	1		3	–
Π_{i1}	9	–	Π_{i2}	9	1	Π_{i3}	9	1
	10	–		10	–		10	1
Δ_{i1}	6	1	Δ_{i2}	6	–	Δ_{i3}	6	2
	7	2		7	2		7	4
	8	1		8	6		8	2

is enough or the social impact's weight is the highest one, all demands can be fulfilled. Therefore, weighting objective functions are the mirror of the priorities of decision-makers, and it must be precise to get the most suitable results and recommend the best network suited to the problem.

Some important managerial conclusions according to the numerical outcomes are presented as follows:

- A mathematical model is proposed to design an SRRSCN that provides an appropriate insight for the decision-makers to manage the SCN during pandemics considering the three main features, sustainability, resiliency, responsiveness, and correlation between them. This research can be valuable for SC managers to witness the impact of uncertainties on SCN.
- Concerning the impact of RMCGP weights, Figs. 6 to 8 indicate that increasing the weights leads to reducing objective functions. The higher weights for the minimizing social negative impact objective function have some correlations with third objective function optimization. In real-world conditions, SC's managers can utilize some strategies to consider weights based on cost reduction. For instance, a guideline that can be helpful is importing PCR kits in the first variant of a pandemic to manage the unpredictable demands. Considering balanced weight social and environmental aspects correlation can affect positively. While this strategy can cause some environmental footprint or extra costs, it can prevent the significant effect of shortage and save people's life.
- As indicated in Fig. 9, priority of met-demand rate can change in different variants of the pandemic. After global vaccination, the demand satisfaction rate stayed at the minimum amount because costs are important at this time. In the real-world challenge, when the SC manager passed the challenging time, then they can focus on cost reduction and reduce the import. Instead, they can provide the kits just with the minimum demand satisfaction and consider transportation types with lower carbon footprints with increasing the weight of the third objective function.

7. Conclusion and future directions

In summary, this article proposes a sustainable-resilient-responsive supply chain network model to address the challenges of designing an effective supply chain for critical products during a pandemic. The proposed model takes into account three key aspects of sustainability and utilizes a mathematical model to minimize total costs, negative social impacts, and carbon footprint, while also considering backup suppliers to improve resilience against sporadic disruptions. A responsiveness constraint is also defined to ensure a specified level of minimum satisfaction under each scenario to prevent possible shortages.

The research also implements stochastic programming, which is an effective approach to manage uncertainty in supply chain management problems. It provides a way to address random disruptions and sudden

changes in demand by considering different scenarios and optimizing solutions for each scenario. The proposed model uses the RMCGP method, which is a powerful optimization technique, to solve the mathematical model and obtain optimal solutions. The sensitivity analysis conducted on the model's objective functions highlighted the significance of the RMCGP weight and the minimum percentage of demand satisfaction in determining the feasibility and efficiency of the solution. The study results indicate that demand fluctuations can leave some capacities of distribution centers underutilized, while suppliers have the potential to provide additional supplies or vice versa. Consequently, it is crucial to strike a balance between the capacity of suppliers and distribution centers to eliminate capacity bottlenecks. Resource sharing, such as equipment and labor, among different sectors involved in the operations can enhance the overall capacity of supplying goods. Thus, it is imperative to develop strategies that optimize capacity allocation between suppliers and distribution centers while ensuring effective resource sharing among different sectors. Future research can explore innovative techniques such as collaborative planning involving game theory, forecasting, and replenishment to enhance capacity utilization while minimizing costs and improving efficiency in the supply chain network.

The real case study of a COVID-19 supply chain network demonstrated the effectiveness of the proposed model (Mazandaran University of Medical Sciences, 2021). The outcomes provided valuable insights into the optimal values of objective functions for different scenarios, including critical conditions during the pandemic. This information is particularly useful for policymakers and decision-makers who need to plan and manage supply chains during pandemics and other similar situations. Overall, this research provides a valuable framework that can help to ensure the availability and reliability of essential goods during unexpected disruptions, making it a crucial contribution to supply chain management research.

Finally, this study suggests some future research directions that can further enhance the accuracy and effectiveness of the proposed model. One of the key future directions is to assess the proposed model for other types of disasters such as earthquakes and wars, or for any other critical product during future pandemics. This would help to confirm the accuracy and applicability of the model for unpredictable disruptions in different contexts.

An additional future direction that could be pursued is to explore the use of large-scale instances to define more realistic problems and to employ heuristic/meta-heuristic algorithms, such as NSGA-II, to solve the NP-hard problem on larger scales. This approach would enable the model to be implemented in real-world scenarios with greater accuracy and efficiency, thereby enhancing its practical applicability, particularly for finding Pareto-optimal solutions. Moreover, future research could investigate the impact of uncertain facility capacity and delivery lead-times on the supply chain network, which can provide valuable insights for managing the supply chain during disasters.

Finally, flexible location of manufacturers/suppliers can also be considered to send critical products, which can help to mitigate

disruptions caused by the pandemic. This approach could help to identify optimal locations for manufacturers and suppliers based on the dynamic demand and supply conditions.

Credit authorship contribution statement

Amirhossein Moadab and Ghazale Kordi devised the project and did the main investigation and conceptualization of the work. They also worked on methodology, validation, visualization, and analysis. Mohammad Mahdi Paydar, Ali Divsalar, and Mostafa Hajiaghahi-Keshmeli supervised the designing the methodology, findings of this work and they worked on the validation, editing and reviewing of paper. All authors discussed the results and contributed to the final manuscript, writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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